Charlie Johnson

PhD Candidate, Imperial College London · Computational Biologist

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SUMMARY

PhD candidate at **Imperial College London** developing a causal framework for imaging biomarker development in lung cancer, with 5+ years of concurrent industry experience as a Senior Translational Scientist at **Tempus AI**: turning a 300,000+ patient multimodal oncology database into biomarker and clinical-development decisions for 15+ pharma and biotech partners. Combines rigorous causal and statistical methods with the pace and stakeholder demands of industry.

EDUCATION

Imperial College London, Department of Surgery & Cancer — London, UK
PhD Candidate, Clinical Medicine Research — 2025 – Expected 2028

- Thesis: *A Causal Framework for Imaging Biomarker Development for Personalised Lung Cancer Management*
- Developing causal methods for robust lung cancer imaging biomarkers, separating signal from confounders (e.g. stage, PD-L1 status) via deep structural causal models, counterfactual image generation, and heterogeneous treatment-effect estimation on CT/PET imaging from 1,000+ patients.
- First-year outputs: first-author paper at MICCAI 2026; first-author posters at EuroCIM 2026 and AI in Healthcare 2026.
- **Board Member, Imperial Women's PhD Network** (Sep 2026 – Present): organise and host external speaker events across all academic departments.

MS, Biomedical Informatics, NYU Grossman School of Medicine — 2021
BS, Mathematics (Minor: Biology), Georgetown University — 2020

PROFESSIONAL EXPERIENCE

Tempus AI — New York, NY
Senior Translational Scientist I — Sep 2025 – Present
Translational Scientist II — Feb 2023 – Sep 2025
Junior Translational Scientist — Jul 2021 – Feb 2023

- Lead analytics for 15+ pharma and biotech partners across 25+ cancer indications, turning partner goals into focused research plans that inform biomarker selection, indication expansion, and Phase 1/2 trial strategy.
- Analyse a 300,000+ patient multimodal dataset (DNA-seq, RNA-seq, EHR, claims) using survival analysis, regression, differential expression, and unsupervised machine learning.
- Develop methods to correct missing data, selection bias, and measurement bias in real-world data, turning previously unusable records into analysis-ready evidence.
- Built scalable pipelines and internal R packages used across the scientist team and set team coding standards, reducing time-to-insight
- Present findings to scientific and executive audiences; train new hires and client teams; run workshops as a member of the Tempus Board of Education.

Earlier experience

- **Research Analyst, Fenyö Lab, NYU Grossman School of Medicine** (Dec 2020 – Jul 2021): proteogenomic subtyping of endometrial carcinoma (ICA, WGCNA, GSEA). Co-author, *Cancer Cell* 2023.
- **Investment Banking Summer Analyst, Healthcare, BofA Merrill Lynch** (Summers 2018–19); received full-time offer.

SELECTED PUBLICATIONS & PRESENTATIONS

- **Johnson I**, et al. Counterfactual stress testing for fixing lung cancer segmentation. *MICCAI* 2026 (paper).
- **Johnson I**, et al. Causal clinical-radiomics analysis reveals heterogeneous survival benefit from chemotherapy in lung cancer patients. *AI in Healthcare (AIiH)* 2026 (poster).
- **Johnson I**, et al. Interpretable radiomic survival analysis in lung cancer using deep structural causal models. *EuroCIM* 2026 (poster).
- Clinicogenomic landscape of metastatic colorectal cancer patients with pathogenic *GNAS* variants and real-world outcomes on first-line oxaliplatin. *ASCO* 2025 (poster).
- Dou Y, …, **Johnson I**, …, Fenyö D. Proteogenomic insights suggest druggable pathways in endometrial carcinoma. *Cancer Cell* 2023;41(9):1586–1605.

SKILLS

Methods	Causal inference (structural causal models, counterfactuals, heterogeneous treatment effects), radiomics, image segmentation, survival analysis, real-world evidence, biomarker discovery, multi-omics, deep learning
Languages	R (incl. package development), Python (incl. PyRadiomics), SQL, Bash
Platforms	Google Cloud (BigQuery, GCS), AWS, dbt, Docker, Git/GitHub